Prediction and Prediction-Error-Driven Learning in Children: Developmental Patterns in English Verb Bias Acquisition

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Children routinely anticipate upcoming linguistic structure, and theories propose that these predictions are shaped by prior learning and revised when mismatches generate prediction error (Dell & Chang, 2014). Prediction error thus serves as a key mechanism for updating internal representations. Although children can learn from prediction errors (Gambi & Messenger, 2023), little is known about how these processes develop. Given rapid developmental changes in executive function, inhibitory control, and linguistic experiences, both prediction processing and sensitivity to prediction error may vary with age. This study examines how age modulates (a) learning-induced prediction and (b) prediction error-based learning.

Fifty-five English-speaking children (8-12 years; *M* = 10.0; 22 female) completed a verb-bias learning task. During training, children were briefly exposed to originally neutral verbs (e.g., *hug*) that were shifted toward either an instrument or a modifier interpretation of a *with*-phrase (Figure 1). They were then tested on ambiguous *with*-phrase sentences while their eye movements and offline picture-selection responses were recorded, allowing us to assess how newly learned verb biases shaped prepositional phrases interpretation (Figure 2).

To evaluate developmental differences of learning-dependent prediction, we analyzed both offline interpretation behavior and online anticipatory eye movement. Older children showed sharper contrasts between training conditions over time: they were more likely than younger children to choose the target instrument for sentences containing instrument- than modifier-trained verbs (training type x time point x age interaction, $p = .045$) (Figure 3). In contrast, anticipatory looking measures, as measured by first fixation upon hearing the verbs, showed no age differences in the extent to which children learn to predict target instruments. To test how prediction error motivates learning, we test the effect of the pre-training looks or actions that were opposite to the assigned training condition on learning outcomes. Across participants, greater prediction-incongruent behavior at pre-test was associated with stronger learning in training-consistent first looks and offline choices (p's <.05). Although younger and older children did not differ at baseline, the interaction between age and priors inconsistent with training was marginally significant ($p = .074$; $BF_{10} = 1.35$). Numerically, younger children showed a stronger positive association between prediction-incongruent first fixations at pre-test and subsequent learnings in anticipatory looking (Figure 4), whereas this relationship was attenuated in older children. No developmental differences were observed in the effect of prediction errors on offline learning.

Taken together, these findings suggest that learning-based prediction and prediction error-based learning mechanisms undergo developmental changes. Older children showed stronger learning-dependent prediction in their offline interpretations but no corresponding age differences in online anticipatory looks, suggesting that accumulated experience supports stable verb-structure expectations even as moment-to-moment predictive processing remains

consistent across age. In contrast, exploratory trends in the eye-tracking data suggest that younger children may show greater sensitivity to prediction error during online processing; however, the statistical evidence for this age moderation was weak. Findings hint at younger children may rely more heavily on immediate discrepancy signals to update prediction but may be less able to consolidate these updates into stable representations. However, because age differences may be confounded with variation in prior linguistic experience, future work will disentangle developmental changes in predictive processing from differences in pre-existing structural biases when interpreting ambiguous *with*-phrase sentences.

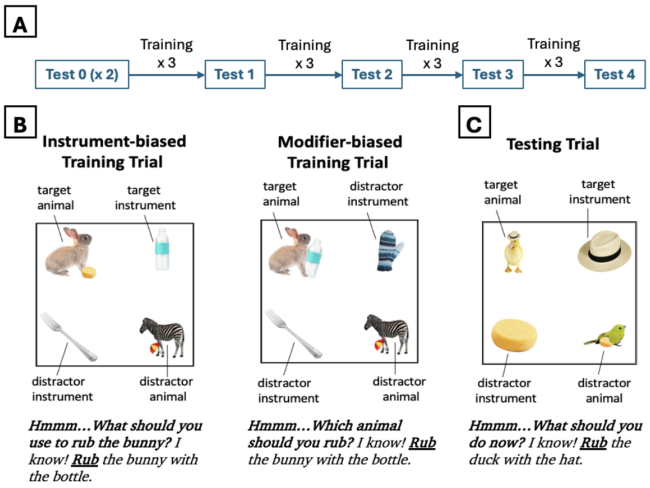

Figure 1. Verb bias learning task design and procedure.
(A) The training and test schedule for each verb
(B) Unambiguous verb bias training trials
(C) Ambiguous verb bias test trials

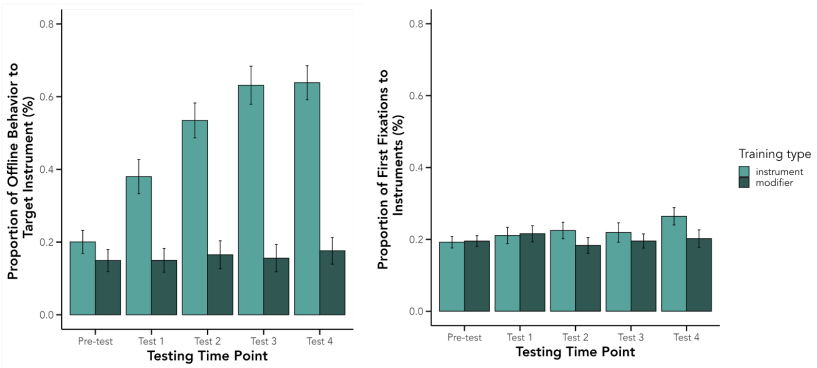

Figure 2. Verb bias learning results
(A) Proportion of offline target instrument choices across five testing time points
(B) Proportion of first fixation to either target or distractor instrument across five testing time points

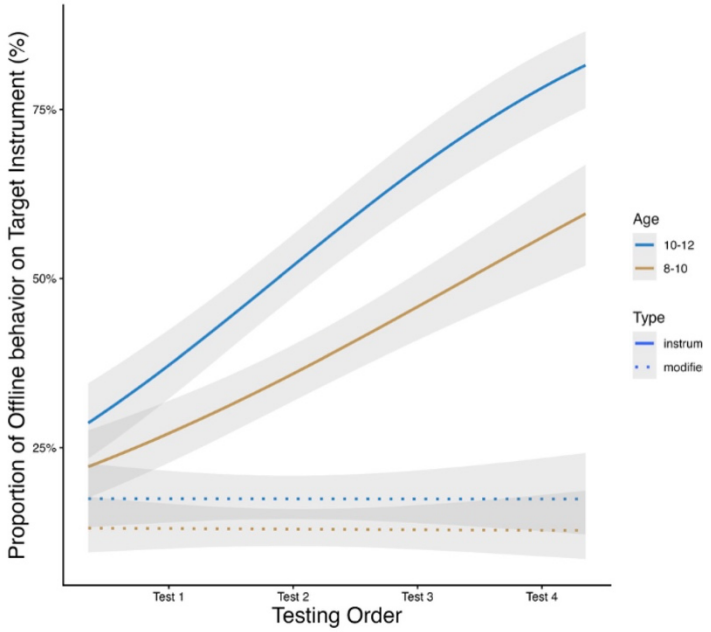

Figure 3. Proportion of offline behavior over the course of verb bias learning

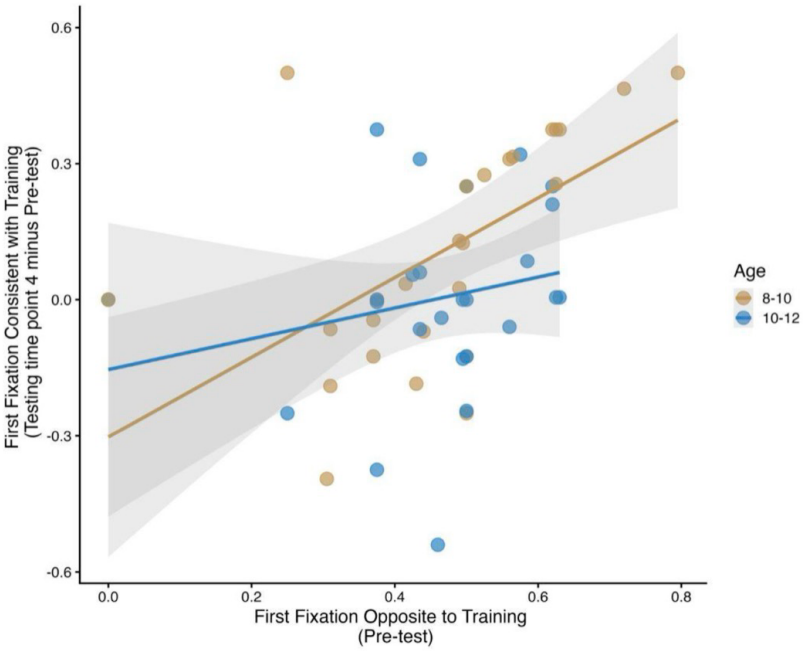

Figure 4. Age moderation on the correlation between individuals' initial verb biases and their learning outcomes in first fixation